

obsidian-mcp-router — quick reference

v0.86.0 · multi-vault Obsidian MCP server, shipped by its Claude Code plugin — one install = everything (51 slash commands · 50 MCP tools · 47 skills)

github.com/tboome33/obsidian-mcp-router

The router in 30 seconds

A single MCP process that knows **all your Obsidian vaults** (local + remote). Each tool takes a `vault` parameter (or uses the default). No more registering one MCP server per vault.

- **One plugin install = everything:** the Claude Code plugin ships *and launches* the server — declared in its root `.mcp.json` (key `router`, via `${CLAUDE_PLUGIN_ROOT}`). A manual `~/.claude.json` entry is the dev alternative; keeping both runs **two** server processes with duplicated tools (`mcp__obsidian-router__*` vs `mcp__plugin_obsidian-router_router__*`)
- **Local + remote vaults** treated identically — drop URL + API key into the config, done
- **Cross-vault:** `vault: "*"` auto-fans-out across every vault
- **Lock mode:** restrict the session to a single vault (safety, focus) · **Auto-enrichment:** Claude suggests wiki saves at 3 natural moments, 4 modes (`ClaudeAsk` / `Hybrid` / `FullAuto` / `off`)
- **Conversion toolbox:** PDF / DOCX / XLSX / PPTX / image (OCR) / audio / YouTube / webpage / git repo → markdown; opt-in **Docling** high-fidelity PDF + `pdf_to_images` (the model sees a PDF page); **BM25 relevance filter** strips off-topic blocks before synthesis
- **Knowledge graph:** typed graph JSON + reading tours + neighbors + shortest link path; export as `llms.txt` or **OKF bundle** (Google's Open Knowledge Format v0.1, with one of the ecosystem's first validators)
- **Wiki analysis:** `find_boundary_pages` ranks heavily-linked-but-thin "frontier" pages; `find_twin_pages` finds quasi-duplicate pairs by cosine over the Smart Connections vectors, with a threshold derived from *your* vault's own distribution — and works on remote vaults too (bridge $\geq 0.9.0$ serves the vector store)
- **Click-to-open:** clickable links in chat via the bridge's `GET /open/<path>` route; `open_in_obsidian` does the same server-side (no browser — ideal for Claude Desktop). The link survives losing the vault's disk (`insecurePort` declarable in the config, exported only on explicit `--with-click-to-open`), and a one-click `GET /ping?v=<vault>` self-test proves *which* vault answers on a port
- **Vault-creation wizard:** `plan_vault` → adjust → `provision_vault`, defaults-first (happy path = 1 interaction)
- **Multi-tenant mode** (opt-in): scope an instance by env vars — vault whitelist, read-only, per-user write audit — for MCPHub/proxy deployments
- **10 hooks:** 2 come active with the plugin for every installer (hot-cache load + decisions recall); the other 8 (session auto-journal, wiki autocommit, link linter, doc-drift checker, update check, hot-cache guard...) are opt-in via `node scripts/setup-vault.mjs --install-hooks`

How the pieces fit — the dependency chain

```
Obsidian ← Local REST API (community plugin) ← BRIDGE (mcp-router-bridge)
  ↑ HTTP per vault (port + apiKey from the Local REST API plugin)
MCP SERVER (obsidian-mcp-router) – Node ≥ 20.19.0 process on the PC
  ↑ MCP over stdio, spawned by Claude Code
CLAUDE CODE PLUGIN (obsidian-router) – commands + skills + agents + hooks
  · the plugin SHIPS and LAUNCHES the server itself
```

- **Bridge** — runs *inside* Obsidian; registers extra routes on Local REST API's HTTP server (`/search/smart`, `/templates/execute`, `/open/*`, `/ping`, `PUT /vault-cas/*` ≥ 0.7.0 — atomic `ifMatch` writes —, `GET /smart-env/sources` ≥ 0.9.0 — serves the Smart Connections vector store, a dot-directory the REST API itself won't serve, so `find_twin_pages` works on remote vaults —, presence heartbeat). **Optional per vault:** without it, core file CRUD still works (plain Local REST API routes), but smart search, Templater execution, click-to-open links, atomic conditional writes and remote twin-page analysis don't (`ifMatch` then degrades to a checked non-atomic fallback). Updates itself via BRAT from GitHub releases.
- **MCP server** — Node process spawned by Claude Code. Requires the **Local REST API** plugin in every vault (mandatory) and its registry at `~/.claude/obsidian-mcp-router/config.json` (maintained by `setup-vault.mjs`); the bridge is optional per vault (see above).
- **Claude Code plugin** — its commands and skills orchestrate the server's MCP tools; two hooks (hot-cache load, decisions recall) read vault files directly from disk.

The plugin ships the server — installing (or updating) the plugin installs (or updates) the server in the same move; fresh installs self-heal `node_modules` via a zero-dependency `bin/` bootstrapper. No `~/.claude.json` entry needed — that path remains for dev checkouts only (`claude --plugin-dir <checkout>` replaces the installed plugin for a session).

Vault setup

1. Have Node.js ≥ 20.19.0. Conversion backends are opt-in (no npm `postinstall`): `npm run install-markitdown` (needs Python 3.10+ on `PATH`), `npm run install-docling` — until installed, the conversion tools stay listed and return an actionable install hint
2. Check whether that opt-in is done *before* a call fails: every `list_vaults` response carries `conversionToolbox` (`available`, `via`, `verified`, `hint`), and `meta-status` renders it in one line. **Eight** tools stop working without markdown; `youtube_to_markdown` falls back to yt-dlp captions (only where yt-dlp is itself installed), and `git_repo_to_markdown` (repomix) and `pdf_to_markdown_docling` are unaffected. Silence the offer for good with `OBSIDIAN_ROUTER_SKIP_MARKITDOWN=1`.
3. **Easiest path:** `/obsidian-router:meta-attach-vault` — interactive wizard that attaches a vault to a workspace (common case), bootstraps a standalone vault, or registers a remote one. Provisions plugins + scaffolds the wiki + binds `.env` + conventions picker.
4. CLI alternative:

```
npm run setup-vault -- "C:\VAULTS\MyVault"
```

⇒ installs/configures **Local REST API** + bridge plugins, writes `.env`, allocates a unique port, adds to the registry. The 2 base hooks come active with the plugin; the 8 extra hooks are opt-in via `node scripts/setup-vault.mjs --install-hooks`.

5. Optional for `search_smart`: **Smart Connections** + **obsidian-mcp-router-bridge** · for Templater: **Templater**

6. Verify: `/obsidian-router:meta-status` or just say *"diagnose the router"*

Configuration — environment variables (workspace `.env`)

VAULT_PATH	Path of the current vault. Auto-detection: if it matches a registered vault, that vault becomes the default. Written by <code>setup-vault.mjs</code> .
OBSIDIAN_ROUTER_DEFAULT_VAULT	Explicit override of the default vault for this process. Wins over <code>VAULT_PATH</code> and over <code>config.defaultVault</code> . Also the workspace↔vault binding used by the wiki-query-first hooks.
OBSIDIAN_ROUTER_LOCKED	Locks the router to one vault for the session. Any tool call to another vault throws. Set via <code>/obsidian-router:lock <vault> --persist</code> .
OBSIDIAN_ROUTER_AUTO_ENRICH	Wiki auto-enrichment mode: <code>ClaudeAsk</code> (default) <code>Hybrid</code> <code>FullAuto</code> <code>off</code> . Set via <code>/obsidian-router:auto-mode <Mode> --persist</code> .
OBSIDIAN_ROUTER_ALLOWED_VAULTS	Multi-tenant: comma-separated whitelist of vaults this instance sees. Others are skipped.
OBSIDIAN_ROUTER_READONLY	Multi-tenant: truthy (<code>1</code> / <code>true</code> / <code>yes</code> / <code>on</code>) hides the 15 write tools — <code>write_file</code> , <code>append_to_file</code> , <code>patch_file</code> , <code>set_frontmatter</code> , <code>merge_frontmatter</code> , <code>move_file</code> , <code>delete_file</code> , <code>execute_template</code> , <code>download_page_assets</code> , <code>build_wiki_graph</code> , <code>provision_vault</code> , <code>write_bundle</code> , <code>record_source</code> , <code>refresh_okf_projections</code> , <code>build_search_index</code> — filtered from ListTools AND refused at call time.
OBSIDIAN_ROUTER_USER_ID	Multi-tenant: audit trail — every successful write appends an attributed line to the vault's <code>wiki-meta/journal.md</code> .
VAULT_<NAME>	A whole vault defined in one env var (JSON: <code>name</code> , <code>baseUrl</code> , <code>apiKey</code> ...) — dashboard-editable on MCPHub deployments.
OBSIDIAN_ROUTER_NO_WATCH	Disables hot-reload of the config file (useful for debugging).

OBSIDIAN_ROUTER_NO_HOT_CACHE_LOAD
• ...NO_DECISIONS_RECALL • ...
NO_WIKI_AUTOCOMMIT • ...
NO_SESSION_JOURNAL

Per-hook opt-outs: disable the hot-cache load, decisions recall, wiki autocommit and session journal hooks respectively. All accept `1` / `true` / `yes` / `on` .

Important files — where things live

~/.claude/obsidian-mcp-router/config.json

Main router config (vaults, port registry, defaultVault, disabledVaults, remoteVaults). Hot-reload enabled.

~/.claude.json

Manual/dev setup only: a hand-registered server entry pointing to a checkout. The plugin declares its server in its own root `.mcp.json` (key `router` , `${CLAUDE_PLUGIN_ROOT}`); a hand-registered entry *plus* the plugin = two processes with duplicated tools (dev machines may keep both deliberately).

~/.claude/settings.json

The 8 opt-in hooks land here via `setup-vault.mjs --install-hooks` (the 2 base hooks ship active in the plugin's `hooks/hooks.json` — 10 hooks total). The plugin itself is enabled **per-workspace** via `<workspace>/.claude/settings.json` — 51 commands + 47 skills ≈ 10k context tokens, only load them where you use Obsidian.

<workspace>/.env

Environment variables for the current workspace (above). Auto-loaded at router boot.

<vault>/CLAUDE.md

Vault wiki consigne — loaded by Claude Code into context. Includes the auto-enrichment consigne (4 modes) + installed conventions.

<vault>/wiki-meta/{catalog,journal,hot,overview}.md

Karpathy LLM-wiki scaffolds (catalog, append-only journal, hot cache, overview) — under `wiki-meta/` (user pages live under `wiki/`). Named `catalog` / `journal` rather than `index` / `log` because OKF reserves those two basenames; the old names are still read. Scaffolded by `/obsidian-router:wiki` .

<vault>/wiki-meta/Sessions/

Per-session detail journals (auto-written by the `session-auto-journal` hook; `journal.md` stays a thin index).

obsidian-mcp-router/docs/

`auto-enrichment.md` (4 modes + 4 placement channels), `features/` (13 prose feature pages), `architecture.md` , this PDF (EN + FR).

51 slash commands /obsidian-router:<name> · auto-trigger on natural language

(EN + FR)

 17 MCP wrappers — one per core vault tool

Categories: discover · read · write · manage · template · convert

COMMAND	EFFECT	TRIGGER PHRASE (EN/FR)
discover-list-vaults	List vaults (online/latency/lock state)	"list my vaults" / "liste mes vaults"
discover-list-files	List files in a vault directory	"list files in Sessions" / "liste les fichiers de X"
read-get	Read a file (markdown + frontmatter)	"show me X" / "montre-moi X"
read-search	Plain-text (substring) search	"grep for <text>" / "trouve <texte>"
read-search-smart	Semantic search (Smart Connections)	"find notes about X" / "trouve mes notes sur X"
read-frontmatter	Read frontmatter (object or one key)	"metadata of X" / "quel est le statut de X"
write-create-or-replace	PUT — create or replace a file	"save this as X.md" / "crée une note X"
write-append	POST — append to a file	"append to my journal" / "ajoute à X"
write-patch	Surgical PATCH (heading/block/frontmatter)	"edit X section in Y" / "édite la section X dans Y"
write-frontmatter-set	Set one frontmatter key	"tag this with X" / "passe le statut de X à closed"
write-frontmatter-merge	Set multiple keys in sequence	"on X set status=closed outcome=tp1"
write-bundle	Multi-file bundle — all-or-nothing apply with journaled rollback	"write these 4 pages atomically" / "écris ces pages d'un bloc"
manage-move	Move or rename a file	"move X to Y" / "renomme X en Y"

manage-delete	Delete a file (2-step confirm guard)	"delete X" then "yes confirm=true"
template-execute	Execute a Templater template	"run X template" / "rends Templates/X.md avec arg=v"
pdf-to-markdown	Local PDF → markdown (MarkItDown, fast plain-text)	"convert this PDF to markdown" / "convertis ce PDF"
pdf-to-markdown-docling	High-fidelity PDF → markdown (Docling, tables/layout, opt-in)	"convert with docling" / "conversion haute fidélité"

3 router-state commands (lock + auto-enrichment)

COMMAND	EFFECT	TRIGGER PHRASE (EN/FR)
lock	Restrict the session to one vault (volatile or <code>--persist</code>)	"I only want to work on X" / "verrouille sur X"
unlock	Lift the lock, restore multi-vault routing	"unlock vaults" / "déverrouille les vaults"
auto-mode	Set the auto-enrichment mode (ClaudeAsk / Hybrid / FullAuto / off)	"switch to Hybrid mode" / "passe en mode Hybrid"

7 conversational helpers (setup + diagnostic)

COMMAND	EFFECT	TRIGGER PHRASE (EN/FR)
meta-setup	Manual/dev install (clone + npm link + <code>~/ .claude.json</code> entry) — normal installs get the server from the plugin	"setup obsidian-mcp-router" / "installe le router"
meta-attach-vault	Wizard — attach a vault to a workspace (common), standalone, or remote	"attach a vault to this workspace" / "connecte mon vault distant"
meta-status	Health-check every vault with fix hints	"are my vaults reachable" / "diagnostique le router"
meta-sync-template	Propagate the reference vault's plugins/snippets/docs to vaults (picker)	"sync the template to all vaults" / "synchronise le template"
sync-from-github	Update one vault or the whole fleet straight from the GitHub skeleton (plugins, themes,	"sync my vaults from github" / "mets à jour depuis GitHub"

snippets, docs) — no local dev
repo needed

meta-audit-bridge-readiness	Audit click-to-open readiness (bridge, REST API, HTTP, live /open probe)	"is click-to-open ready" / "audite le bridge"
conventions	Install/remove/status/propagate CLAUDE.md conventions cross-vaults	"install source-type on X" / "installe la convention X"



24 knowledge-management commands (Karpathy-style LLM-wiki)

COMMAND	EFFECT	TRIGGER PHRASE (EN/FR)
wiki	Scaffold the wiki in a vault (index, log, hot, overview)	"set up a wiki" / "scaffold un wiki"
wiki-ingest	Ingest a source → entity/concept pages + cross-refs	"absorb this article" / "ingère cette URL"
wiki-query	3-tier RAG over the wiki (no web)	"what does my wiki say about X" / "d'après mes notes ..."
wiki-lint	Health check (orphans, dead links, index drift)	"audit my wiki" / "lint le wiki"
wiki-fold	Idempotent log rollup under wiki/folds/	"fold the log" / "compacte le journal"
hot-compact	Compact an oversized hot.md back to its cache contract (verified backup first)	"compact the hot cache" / "compacte le hot"
save	File the conversation as a typed wiki note (session/decision/ADR/...)	"save this" / "sauvegarde ça"
decision-consolidate	Compress a settled decision page to its essentials; move the full deliberation history to a verified archive note	"consolidate this decision" / "consolide cette décision"
autoresearch	Autonomous web→synth→file loop	"research X online" / "fais une recherche web sur X"
canvas	Create/edit Obsidian .canvas files	"create a canvas for X" / "crée un canvas pour X"
defuddle	Strip noise from web pages before ingestion	"defuddle <url>" / "nettoie cette page"
obsidian-bases	Create/edit Obsidian .base files (database views)	"create a base for X" / "crée une base pour X"
wiki-graph	Build the typed knowledge-graph JSON → <code>wiki-meta/graph/</code> + an <code>.understand-anything/</code> copy for the Understand-Anything dashboard (deterministic, no LLM)	"build the wiki graph" / "construis le graphe"

wiki-tour	Ordered pedagogical reading tour from the link topology	"give me a tour of this vault" / "par où je commence"
wiki-neighbors	One page's neighbors — links out, backlinks, or both	"what links to X" / "voisins de X"
wiki-path	Shortest link chain between two pages	"how is X connected to Y" / "quel rapport entre X et Y"
wiki-export	Export the vault as llms.txt / llms-full.txt or an OKF bundle	"export the wiki as llms.txt" / "exporte le wiki"
okf-export	Export a wiki subset as a shareable OKF v0.1 bundle (conformance self-checked)	"export this folder as an OKF bundle" / "publie en bundle OKF"
okf-projections	Regenerate the generated OKF navigation in wiki/ (root + per-directory index.md, newest-first log.md); auto-refreshed after writes; --check = drift report	"refresh the OKF projections" / "régénère les index du wiki"
okf-check	Validate any OKF bundle against the v0.1 conformance rules	"validate this OKF bundle" / "ce bundle est-il conforme ?"
wiki-refresh-digests	Regenerate per-page digest sidecars (concepts/claims/keywords)	"refresh the digests" / "rafraîchis les digests"
build-search-index	Build/refresh the local BM25 search index (plugin-free search tier, idempotent)	"build the search index" / "construis l'index de recherche"
wiki-boundary	Rank heavily-linked-but-thin "frontier" pages worth writing next	"what should I write next" / "pages frontière du wiki"
who-is-speaking	Identify the family member in a shared vault, lock routing per member	"who is speaking" / "c'est Karine"

The 50 MCP tools — what Claude actually calls under the hood

`list_vaults` · `list_files`

Discovery. Vault catalog with online status + latency (call first); directory listing.

`get_file` · `search` · `search_smart` ·
`get_frontmatter`

Reads. Full file (returns `contentSha256` — the `ifMatch` token); substring search; semantic search (Smart Connections embeddings, cosine scores); typed frontmatter. `vault: "*" = cross-vault fan-out`. `search_smart` tells you two things about its own answer. `freshness` flags each hit whose page has been modified since it was indexed (`changed` / `touched`) or has gone (`page-missing`) — open the page before quoting such a hit; a vault with no disk here answers `checkable: false` rather than implying currency. `folderExclusion` reports what was left out: **by default** `wiki-meta/Sessions` (chronological logs, 41.6% of the indexed pages on this fleet). Pass `excludeFolders: []` to get them back.

`write_file` · `append_to_file` · `patch_file` ·
`delete_file` · `set_frontmatter` ·
`merge_frontmatter` · `move_file`

Writes & file management. Create/replace; append; surgical patch by heading/block/frontmatter; guarded delete (`confirm: true`); one or many frontmatter keys; move/rename. `ifMatch` : replay `get_file` 's `contentSha256` and the write is refused (409) if the file changed since you read it — atomic with bridge $\geq 0.7.0$, checked fallback otherwise.

`execute_template`

Templater. Render a template, optionally write the result; args via `tp.mcpTools.prompt("key")` .

`lock_vault` · `unlock_vaults` ·
`set_auto_enrich_mode`

Router state. Single-vault isolation; auto-enrichment mode switch.

`plan_vault` · `provision_vault`

Vault provisioning. Read-only plan (defaults + questionnaire + warnings) → one-call creation (plugins, port, API key, wiki scaffold, registry). Local-only.

`pdf/docx/xlsx/pptx/image/audio_to_markdown`

Local conversion (Markdown — opt-in via `npm run install-markitdown` ; the tools stay listed and return an install hint when the backend is missing). Office/PDF/image OCR/audio transcription → markdown text; chain with `write_file` .

`pdf_to_markdown_docling` · `pdf_to_images`

High-fidelity PDF (opt-in Docling): layout + table-structure recognition. `pdf_to_images` renders pages to PNG so the model can see them (capped pages/bytes).

youtube/bing_search/webpage_to_markdown ·
git_repo_to_markdown

Remote conversion. URL → markdown (SSRF-guarded); YouTube transcripts; git repo → single markdown bundle (repomix, optional Tree-sitter compression).
`webpage_to_markdown` accepts `relevanceQuery` (BM25) and `citations: true` — inline links move into numbered footnotes with a reference list, one per destination, leaving code, images and wikilinks alone. With both, the filter runs first, so markers and definitions match one-to-one.

filter_relevant_blocks

Relevance filter. Deterministic BM25 second pass over markdown you already have — drops off-topic blocks, keeps frontmatter/headings, multiple no-op safety nets. No fetch, no LLM.

extract_page_metadata ·
propose_linked_sources ·
download_page_assets

Web ingestion support. Deterministic page metadata (JSON-LD/OpenGraph); scored link-following candidates; image download into the vault.

get_wiki_context_pack · build_wiki_graph ·
build_wiki_tour · get_page_neighbors ·
wiki_path

Context & graph. Structured context envelope for external agents; typed knowledge graph; reading tours; per-page neighbors; shortest link path. Every item of the context pack carries its `source` — `index` (the vault's own catalog) and `graph` (a wikilink) are **navigation, authoritative**; `semantic` is augmentation and never the sole support for a factual claim. A pack with no navigational anchor says so: `answer-relies-on-semantic-only`.

find_boundary_pages · find_twin_pages

Wiki analysis. Frontier pages worth writing next (linked a lot, say little); quasi-twin pairs by cosine over the Smart Connections vectors, threshold derived from the vault's own distribution, every pair carrying the evidence to dismiss it. Both answer `available: false` with a reason (never a fake "0 found") when they cannot look. `find_twin_pages` works on **remote vaults** too (bridge ≥ 0.9.0 serves the vector store).

write_bundle · refresh_okf_projections ·
build_search_index

Wiki maintenance. Journaled multi-file bundle — all-or-nothing apply with rollback; regenerate the OKF navigation (per-directory `index.md` + newest-first `log.md` — generated files, never edit by hand); local BM25 search index (works on every vault, no plugin needed).

record_source · audit_sources

Source ledger. Record where ingested content came from (URL, hash, date); audit a vault's sources for drift and gaps.

`build_open_link` · `open_in_obsidian` ·
`get_view_link`

Navigation. Ready-to-paste click-to-open links (path-verified against the vault; on a diskless vault the link is still built, flagged `pathVerified: false`); server-side open (raise Obsidian window, optional heading anchor) — no browser round-trip; ephemeral browser view-link to a live GUI for remote deployments.

Tips · Type `/obsidian-router:` in Claude Code → autocomplete · Every command accepts a `vault` argument (wrappers fall back to the default) · `vault: "*" on search / search_smart` = cross-vault fan-out · Trigger phrases are indicative — Claude recognizes variants · Write results carry a ready-made `clickToOpenUrl` — one click opens the note in Obsidian · Full docs: `README.md`, `docs/features/` (13 pages), `docs/auto-enrichment.md` in the repo.